Rungta College of Engineering and Technology Bhilai (CG)

Department of CSE- Cyber Security

Project Proposal Presentation

OSINT TOOLKIT DASHBOARD

Cyber Security

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OSINT Toolkit Dashboard

Cybersecurity Reconnaissance & Domain Intelligence Platform
WHOIS + DNS Intelligence Dashboard
Built using HTML, CSS, JavaScript, Node.js and Express.js

Introduction

OSINT stands for Open Source Intelligence.
The project analyzes publicly available domain information.

Integrates WHOIS, DNS and Social OSINT reconnaissance tools into one dashboard.

Designed for cybersecurity learning and reconnaissance analysis.

Problem Statement

Traditional OSINT tools are difficult for beginners.

Most tools focus on only one reconnaissance capability.

This project creates a centralized visual dashboard for intelligence gathering.

Objectives

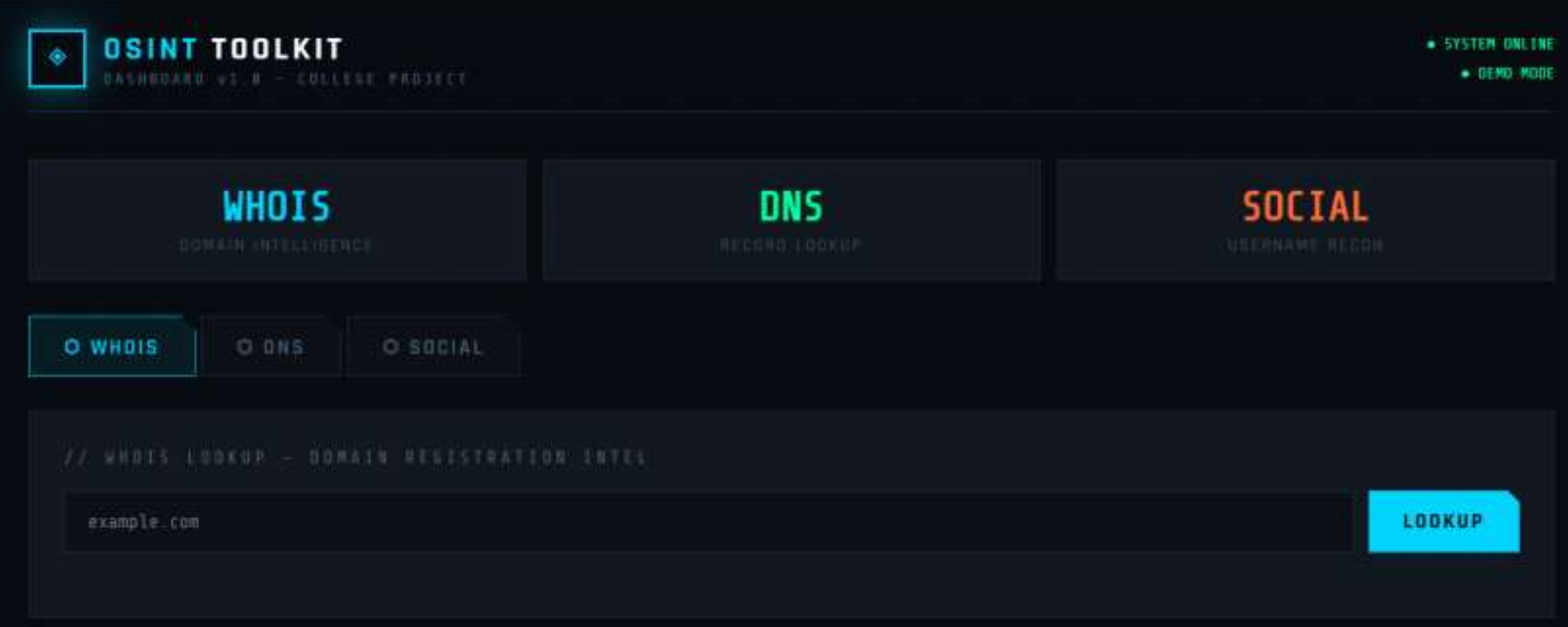
Learn frontend/backend integration.

Implement WHOIS and DNS lookup modules.

Understand API communication using Fetch API.

Build a responsive cybersecurity dashboard interface.

Project Overview



Unified dashboard for domain reconnaissance. Supports WHOIS analysis and DNS record lookups. Frontend communicates with backend APIs in real time.

Technologies Used

Frontend: HTML, CSS, JavaScript

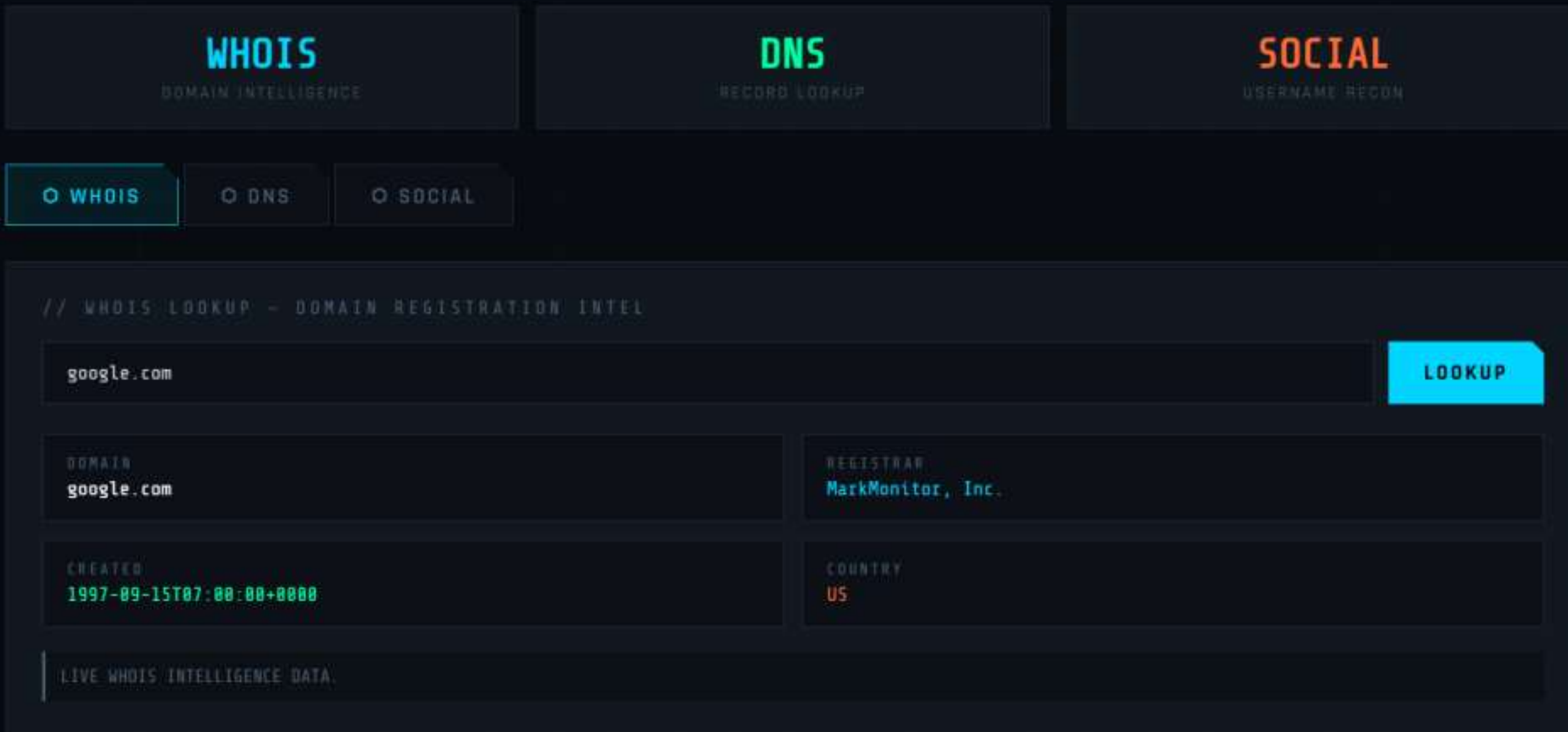
Backend: Node.js, Express.js

Libraries: whois-json, dns, cors

Development Tools: VS Code, Live Server

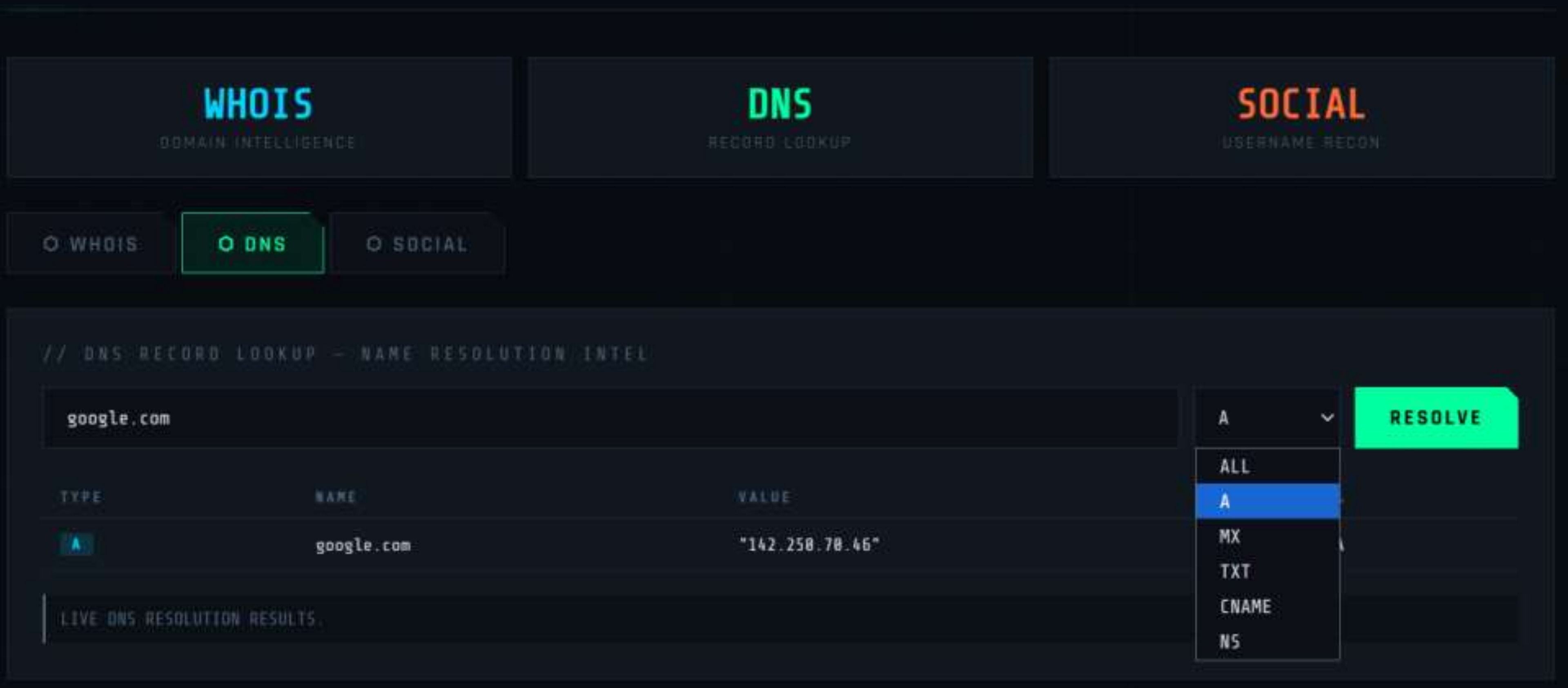
Communication: Fetch API + JSON

WHOIS Module



Performs domain registration intelligence gathering.
Displays registrar, creation date, and country information.
Uses whois-json npm package for backend processing.

DNS Module

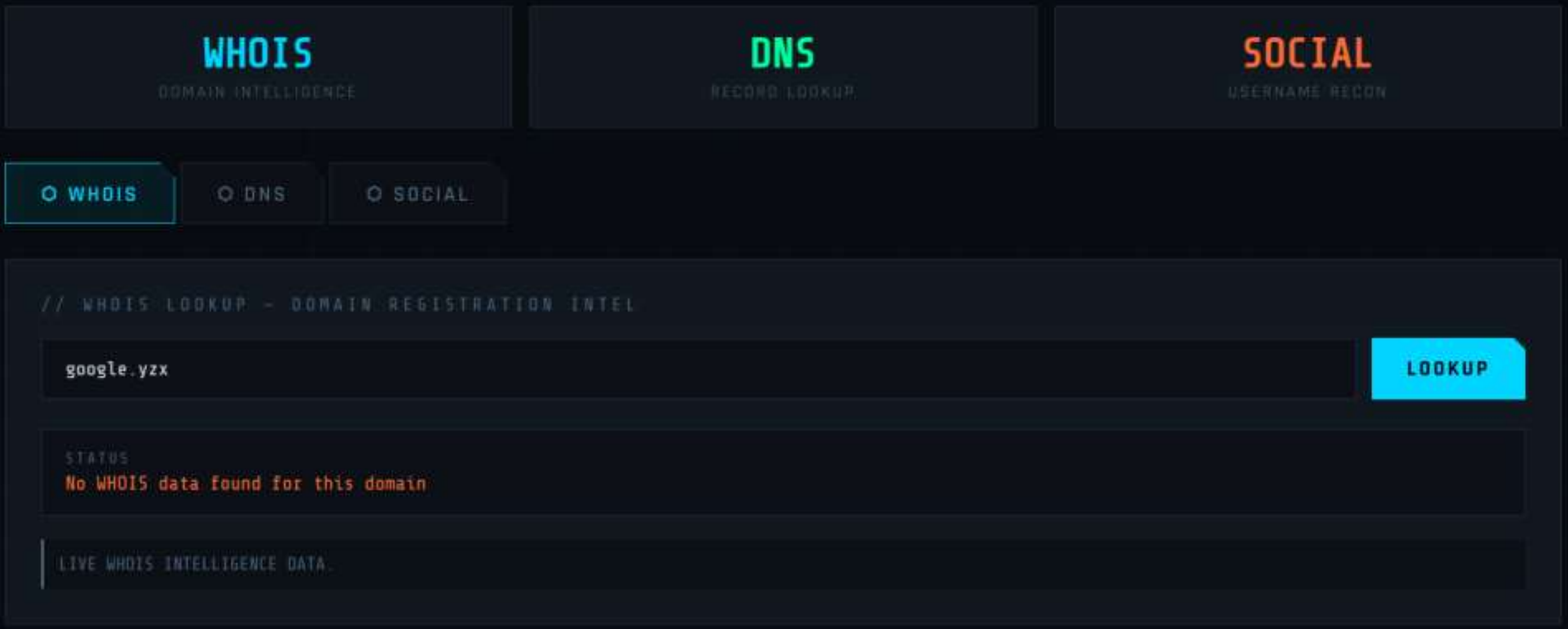


Supports A, MX, TXT, NS and CNAME records.
Displays DNS intelligence information dynamically.
Uses Node.js DNS module for record resolution.

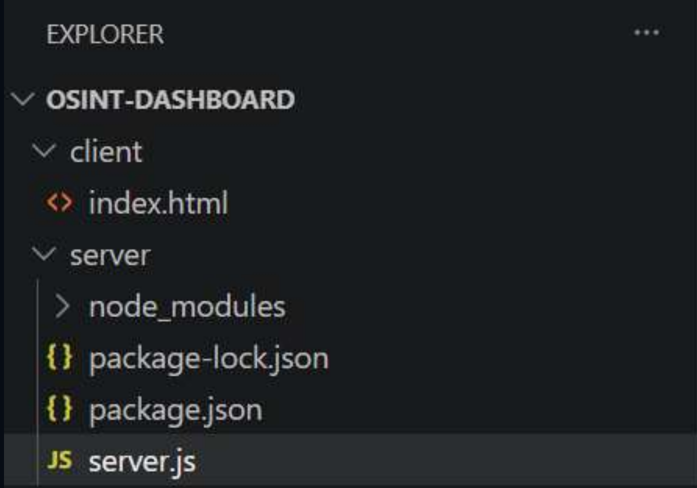
Advanced DNS Analysis

MX records provide mail server intelligence.
TXT records contain verification and security information.
Useful for reconnaissance and infrastructure analysis.

Error Handling & Validation

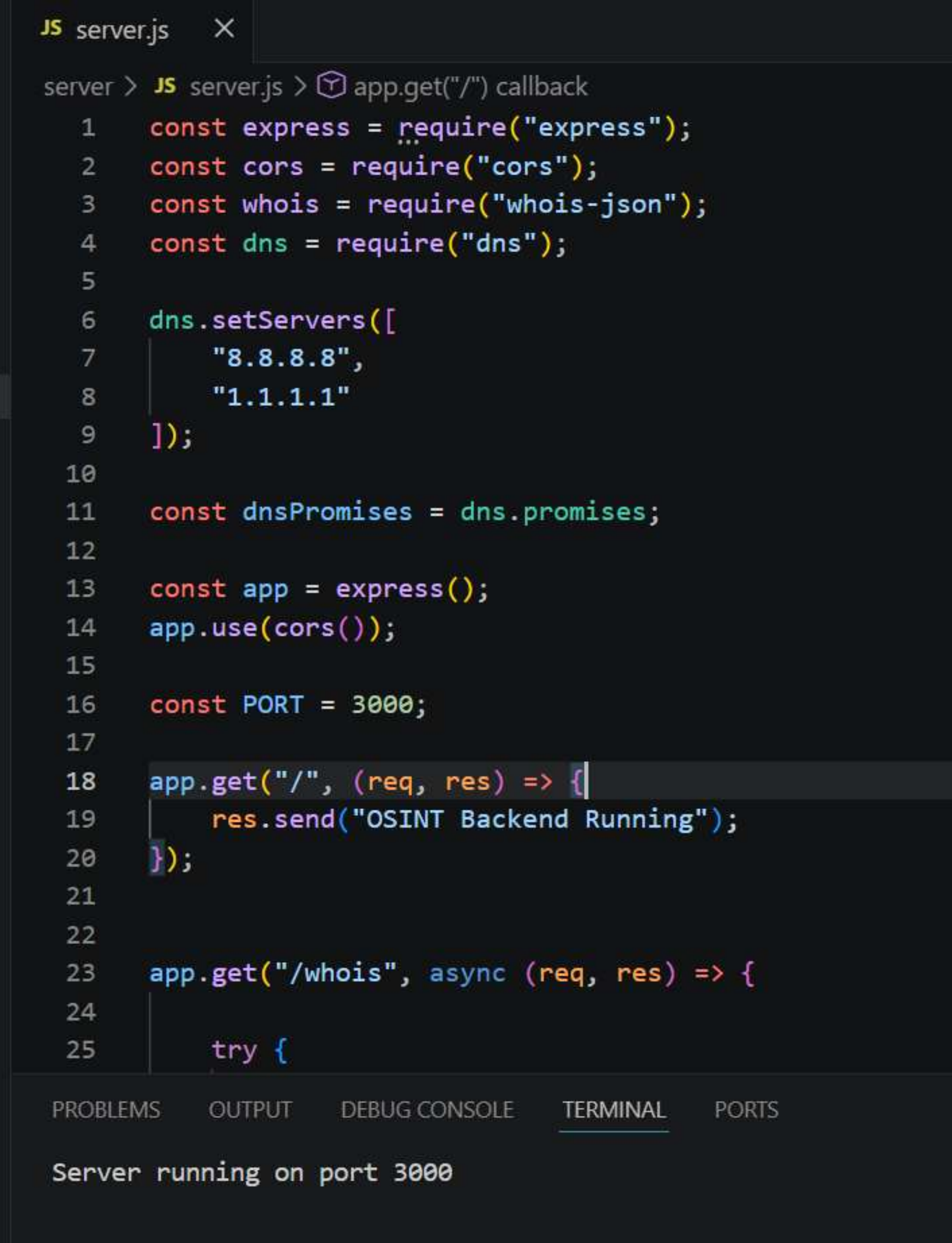


Handles invalid domains gracefully.
Displays cyber-themed error messages inside the UI.
Prevents crashes and improves user experience.



Backend Architecture

Express.js handles API routing.
Frontend communicates using Fetch API.
Backend returns structured JSON responses.
WHOIS and DNS processing occurs server-side.



Workflow / Methodology

User enters domain name
Frontend JavaScript triggers fetch request

Node.js backend receives API request
WHOIS/DNS lookup is processed

JSON response is returned
Frontend dynamically renders results

Future Scope

- Social OSINT integration
- Sherlock username reconnaissance support
- Threat intelligence integration
- Advanced reconnaissance modules

Conclusion

- The project successfully integrates WHOIS and DNS intelligence modules.
- Demonstrates practical understanding of APIs and networking.
- Provides hands-on experience in cybersecurity-focused full-stack development.

Thank You